MASA HACKATHON
Submission Report

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1.0 Problem Statement: Structural Climate Risk in the Reinsurance Layer

The global reinsurance industry is operating on a risk landscape that historical loss data can no longer adequately capture. As greenhouse gas (GHG) emissions accumulate, the physical environment in which insured assets operate is shifting, systematically changing the frequency, severity and geographic distribution of natural disaster losses that flow into the reinsurance layer.

The scientific basis is established. The IPCC AR6 confirms that anthropogenic GHG emissions have unequivocally caused global warming, which intensifies precipitation extremes through the Clausius–Clapeyron relationship and amplifies tropical cyclone intensity in basins directly relevant to Southeast Asia. These are not distant projections but they are already observable in the loss record.

This study focuses on **Thailand and Viet Nam**, selected for their contrasting but complementary catastrophe risk profiles. Thailand's 2011 floods which is one of the costliest insured natural disasters in history demonstrated how concentrated industrial exposure in a flood-prone basin generates catastrophic losses that cascade through the reinsurance layer. Viet Nam's compound vulnerability, with dense coastal populations and economic assets concentrated in low-lying deltas, reflects a high-frequency storm and flood profile with a growing protection gap.

Crucially, the pathway from GHG to reinsurance claims is not linear:

GHG Emissions → Atmospheric Warming → Intensified Hazards → Physical Damage → Insured Losses → Reinsurance Claims

At each stage, real-world frictions, insurance penetration, asset concentration, retention thresholds mediate the outcome. This study explicitly models the upstream climate-pressure signal and traces its implications through reinsurance pricing and accumulation strategy, offering a more physically grounded framework than backward-looking loss histories alone can provide.

2.1 Target Variable: GHG emissions intensity

The modelling target is GHG emissions intensity — total greenhouse gas emissions excluding LULUCF per billion USD GDP. Normalising by economic output controls for the sizable scale difference between Thailand and Viet Nam, enabling a fair comparison of decarbonisation progress. Intensity reveals whether each economy is becoming cleaner per unit of output, whereas absolute emissions obscure this by construction. LULUCF is excluded because land-use fluxes are volatile, methodologically inconsistent and driven by one-off policies rather than structural economic activity; omitting them isolates stable, internationally comparable energy and industrial emissions. For reinsurance assessment, this provides a more robust upstream climate-pressure signal.

2.2 Selected Indicators & Rationale

Indicators were selected against three criteria: (1) mechanistic link to GHG emissions or climate-related insurance loss. (2) adequate data coverage for Thailand and Viet Nam across 1999–2024. (3) structural non-redundancy, with each driver capturing a distinct economic or biophysical transmission channel. Five structural drivers were retained, alongside country fixed effects and a year trend to control for time-invariant country differences and long-run global shocks respectively.

- **GDP per capita (NY.GDP.PCAP.KD)** – Measures average income and overall development. Wealthier economies typically hold larger stocks of buildings,

infrastructure and industrial assets, which increases energy demand and expands the pool of insured property exposed to climate hazards.

- **Renewable energy consumption (EG.FEC.RNEW.ZS)** – Indicates the share of low-carbon energy in total use. A rising share suggests the economy is shifting away from fossil fuels and reducing its long-term emissions trajectory.
- **Energy intensity (EG.EGY.PRIM.PP.KD)** – Records how much primary energy is required to produce one unit of GDP. When intensity falls, the economy is becoming more efficient. This helps separate sustainable growth from expansion driven by heavy, emission-intensive industry.
- **Forest area (AG.LND.FRST.ZS)** – Forests act as carbon sinks and provide protection against natural disasters. In Viet Nam, coastal mangroves reduce damage from typhoon storm surges; in Thailand, upland forests influence river flood patterns. This variable therefore captures both emissions absorption and physical vulnerability.
- **Manufacturing value added (NV.IND.MANF.ZS)** – Measures the importance of manufacturing in the economy. Manufacturing is typically energy-intensive and relies heavily on fossil fuels in emerging markets, raising emissions. It also concentrates valuable industrial property in hazard-prone areas.

2.3 Fixed Effects & Trend Controls

- **Country fixed effects** – The study focuses on Thailand and Viet Nam only. These effects account for fixed differences between the two countries, such as geography and legal systems, so the model can isolate how yearly changes within each country drive emissions.
- **Year trend** – Covering 1999 to 2024, this controls for gradual global events such as the 2008 financial crisis, the 2015 Paris Agreement and the COVID-19 pandemic that affected both countries. This ensures the other indicators capture genuine relationships rather than shared long-term trends.

3.0 Data cleaning and data preprocessing

Initial preprocessing involved treating all columns as text to safely handle the World Bank's ".." notation for missing values, which were subsequently converted to standard NA values. Metadata footer rows were removed, redundant columns (e.g., Time Code) were dropped, and the Time variable was converted to an integer year format. All indicator columns were coerced to numeric, and column names were standardized into a clean snake_case format, with bracketed WDI codes removed for readability.

To ensure a high-fidelity comparative analysis, the dataset was specifically filtered to include only Thailand and Viet Nam for the period 1999–2024. This focused approach eliminated noise from other regional entities and allowed for a precise assessment of the variables relevant to these two nations. Additionally, land area data were integrated from a supplementary source, with 2024 values imputed from 2023 to maintain continuity in the most recent observation year.

A threshold-based imputation strategy was then applied to address remaining missing values within each country group to prevent data leakage. Indicators with more than 40% missing data were dropped entirely to avoid introducing bias. For indicators with 20–40% missingness, a "down-up" filling approach was used, utilizing Last Observation Carried Forward (LOCF) and Next Observation Carried Backward (NOCB). Finally, indicators with less than

20% missingness were imputed via linear interpolation using zoo::na.approx, with any residual edge-case gaps resolved through a final forward and backward fill.

All imputation steps were performed within individual country groups to prevent cross-country data leakage, ensuring that the longitudinal trends for Thailand and Viet Nam remain distinct and authentic.

3.1 Exploratory Data Analysis

Volatility in Year-over-Year (YoY) GHG and GDP Growth

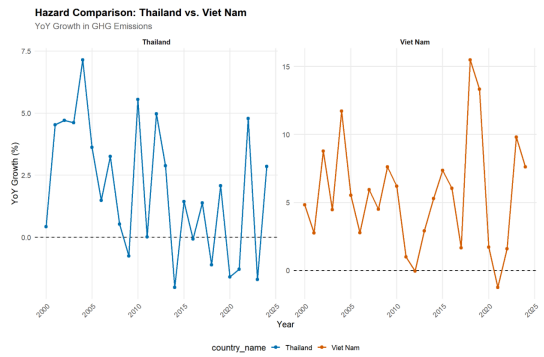


Figure 1: YoY Growth in GHG Emissions

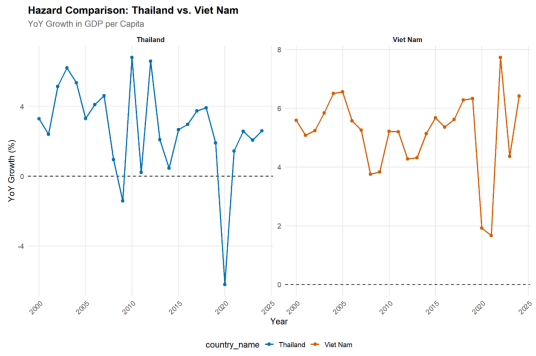


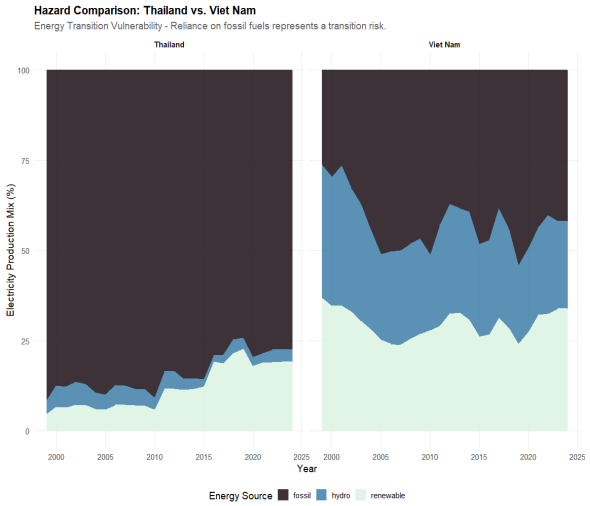
Figure 2: YoY Growth in GDP per Capita

The stability of emission and macroeconomic trajectories directly impacts the predictability of climate-exposed liabilities. Between 2020 and 2024, Thailand exhibited muted GHG volatility, with year-on-year growth fluctuating between approximately -1.60% and 4.78%. Its GDP per capita followed a similarly moderate path, recording a pandemic-induced trough of -6.13% in 2020 before recovering to 2.57% in 2021, 4.40% in 2022, and moderating to 1.97% in 2024. This co-movement suggests that Thailand's emissions pathway is broadly synchronised with predictable macroeconomic cycles, implying lower parameter uncertainty in liability projections.

Viet Nam, however, presents a sharply divergent profile. Its GHG YoY growth showed pronounced structural volatility, surging to 9.80% in 2023 and 7.61% in 2024 after pandemic-era disruptions. Meanwhile, its GDP per capita YoY recovered from 0.60% in 2021 to 5.50% in 2022, then decelerated to 3.65% in 2024. The critical insight is the widening gap between GHG and GDP volatility: Viet Nam's emissions growth is structurally outpacing its economic growth, indicating that industrialisation is becoming more carbon-intensive per unit of output.

Energy Mix Composition and Transition Risk Exposure

Figure 3: Energy Composition Plot

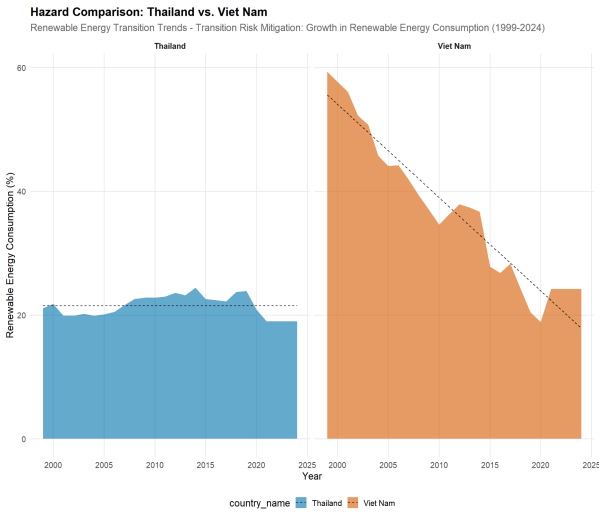


The 2023 energy composition data reveal starkly divergent transition risk exposures. Thailand remains heavily fossil-dependent, with 77.48% of electricity generated from fossil fuels compared to only 19.20% from renewable sources and 3.32% from hydroelectricity. Viet Nam presents a more balanced profile: 41.90% fossil fuels, 24.21% hydro, and 33.89% renewables. Thailand's high fossil dependency concentrates its exposure to carbon pricing shocks, stranded asset risks and abrupt regulatory

transitions. Conversely, Viet Nam's lower fossil share and substantial hydroelectric base provide a partial buffer against transition risk, though hydro dependency introduces physical climate vulnerabilities such as drought-related generation shortfalls.

Renewable Energy Transition Trends

Figure 4: Renewable Energy Trend Plot



This analysis compares renewable energy consumption shares in Thailand and Viet Nam to assess transition risk patterns. Thailand shows a relatively stable but slightly declining renewable share, from 21.8% in 2000 to 19.0% in 2024, indicating limited progress in energy transition and continued reliance on conventional energy sources. Viet Nam exhibits a more volatile trajectory, falling from 57.7% in 2000 to 18.9% in 2020 before recovering to 24.2% in 2024. This reflects a structural shift away from earlier renewable reliance, followed by partial recent recovery. Overall, Viet Nam shows higher transition volatility, while Thailand remains more stable but stagnant, indicating differing speeds and uncertainty in their decarbonisation pathways.

4.0 Comparative Analysis of Thailand and Viet Nam: Natural Disaster Loss Pressure and Reinsurance Implications

4.1 Why Thailand and Viet Nam

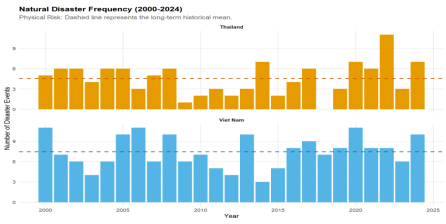
Thailand and Viet Nam are selected because they represent two contrasting Southeast Asian catastrophe-risk profiles. Thailand reflects a low-frequency but high-severity flood accumulation market, where a single major event such as the 2011 flood can dominate the loss experience. Viet Nam reflects a higher-frequency storm and flood exposure market, where repeated disaster events create recurring economic-loss pressure. This contrast allows the analysis to compare two different reinsurance concerns: tail-event severity in Thailand versus aggregate recurring loss pressure in Viet Nam.

4.2 Data limitation and proxy approach

Complete annual insured-claims data is not publicly available in a consistent format for either country. Therefore, EM-DAT economic loss data is used as an open-source proxy for insured-loss pressure, comparing event frequency, annual total economic loss, and average loss per event. To keep the analysis focused on climate-related natural catastrophe risk, the dataset is filtered to include only climatological, hydrological and meteorological disasters, such as droughts, extreme temperatures, floods, landslides, storms and typhoons, while excluding non-climate-related geophysical events. These figures are interpreted as economic-loss pressure, not actual reported insurance claims.

4.3 Frequency Trend: Number of Disaster Events

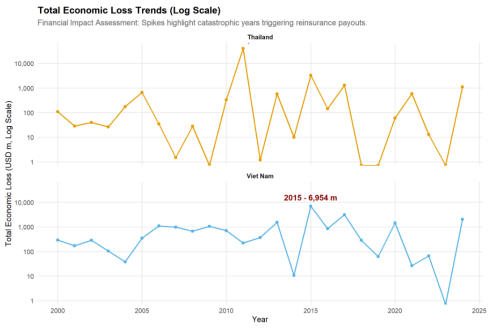
Figure 5: Natural Disaster Event Frequency in Thailand and Viet Nam, 2000–2024



The annual number of disaster events shows that Viet Nam experiences more frequent natural disasters than Thailand across most years. This suggests that Viet Nam has a more persistent disaster-frequency profile, mainly driven by storms, floods and typhoon-related hazards. Thailand records fewer events in many years, but this does not imply lower financial risk because its loss profile is concentrated in occasional high-severity catastrophe events.

4.4 Severity Trend: Annual Total Economic Losses

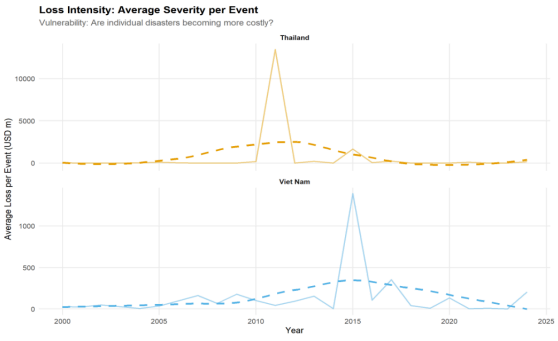
Figure 6: Annual Total Economic Losses from Natural Disasters (2000–2024)



The annual total economic loss trend shows a clear difference between Thailand and Viet Nam. Thailand's losses are catastrophe-spike driven, with the 2011 flood dominating the observation period. This indicates that Thailand's disaster-loss profile is concentrated in low-frequency but very high-severity events. Vietnam also showed a major spike in 2015, with total economic losses of approximately USD6.95 billion, but its losses are not concentrated in one year alone. Several other years also show material economic losses, indicating a more recurring storm and flood loss pattern. Therefore, Thailand reflects a flood-accumulation and tail-severity profile, while Viet Nam reflects a repeated storm and flood economic-loss profile with one notable 2015 severity spike.

4.5 Loss Intensity: Average Loss per Event

Figure 7: Average Economic Loss per Natural Disaster Event (2000–2024)



Average Loss per event =

$$\frac{Total\ Economic\ Damage}{Number\ of\ disaster\ events}$$

The average loss per event metric separates frequency from severity. Thailand shows a sharp severity spike around 2011, confirming that its catastrophe risk is dominated by extreme flood accumulation rather than a steady increase in event count. Viet Nam's average loss per event is generally lower than Thailand's 2011 peak, but it shows a clear severity spike around 2015. Outside this spike, Viet Nam's loss intensity remains more moderate and recurring, suggesting that its risk profile is driven more by repeated storm and flood events than by one dominant extreme event. This means Thailand has stronger tail-event severity risk, while Viet Nam has stronger recurring aggregate loss pressure.

4.6 Insurance claims interpretation

This section does not claim to estimate actual annual insured claims. Instead, it compares disaster-loss pressure and interprets claims relevance using available insured-loss evidence. Economic losses only become insurance claims when affected assets are insured and the loss falls within policy coverage.

Thailand's 2011 flood provides strong, quantifiable evidence of actual insured-loss accumulation because it is widely reported as a major insured-loss catastrophe event. The 2011 floods generated over USD40 billion in economic losses, translating to approximately USD12–15 billion in insured claims (Swiss Re Sigma 2/2012). Conversely, Viet Nam's recurring economic losses suggest a persistent protection gap and underinsurance risk. Because there is limited public insured-loss reporting and limited public evidence that Viet Nam's 2015 economic severity spike translated into a comparable level of insured or reinsured losses, Viet Nam represents a market where physical damage may outpace formal financial risk transfer.

4.7 Reinsurance Implications

Overall, Thailand and Viet Nam show two different catastrophe-risk patterns. Thailand is severity-driven: one major flood can dominate the loss and claims experience, making catastrophe excess-of-loss adequacy, industrial flood accumulation monitoring and tail-risk capital important. Viet Nam is frequency-driven: repeated storm and flood events create recurring economic-loss pressure, but limited public insured-loss evidence suggests a material protection gap. For a regional reinsurer, this means Thailand requires stronger large-event accumulation control, while Viet Nam requires aggregate loss monitoring, pricing adequacy and protection-gap assessment. Historical insured claims alone may understate future exposure, especially if insurance penetration increases and part of today's uninsured economic loss becomes tomorrow's insured and reinsured loss.

5.1 2024 GHG Emissions Intensity Prediction

Three modelling approaches were tested to predict **GHG emissions intensity**, measured as **Mt CO_2e per billion USD GDP**, for 2024. The first model was a **log-linear panel fixed effects model**, which explains log GHG/GDP intensity using renewable energy consumption, energy intensity, forest area, manufacturing share, year, and country fixed effects. The second model was a **dynamic panel fixed effects model**, which extends Model 1 by adding lagged log GHG/GDP intensity. This was included to account for persistence in emissions intensity over time. The third model was a **country-wise ARIMA benchmark**, used to compare whether a pure time-series model could outperform the explanatory panel models. For Thailand, the selected ARIMA model was ARIMA(0,2,1), while for Viet Nam it was ARIMA(0,1,0).

For preprocessing, total population was first reconstructed using population density multiplied by land area. Total GDP was then reconstructed using GDP per capita multiplied by total population. The target variable, **GHG per billion GDP**, was calculated as total GHG emissions divided by total GDP in billion USD. The target variable was log-transformed, and a country-specific lagged log GHG/GDP variable was created to capture persistence while avoiding cross-country leakage. The data were split using a strict time-based approach: 1999–2018 as training data, 2019–2023 as validation data, and 2024 as the final forecast year.

The final selected best model was Model 2 for its best overall validation accuracy, the dynamic panel fixed effects model:

$$\log(GHG/GDP) = \beta_0 + \beta_1 \text{ lag}(\log GHG/GDP) + \beta_2 \text{ Renewable} + \beta_3 \text{ EnergyInt} + \beta_4 \text{ Forest} + \beta_5 \text{ Manufacturing} + \beta_6 \text{ Year} + \text{CountryEffect}$$

Table 1: Model Performance Comparison

Model	RMSE	MAE	MAPE	R^2
Model 1: Panel FE	0.05	0.04	3.60%	0.99
Model 2: Dynamic Panel FE	0.05	0.04	3.32%	0.99
Model 3: ARIMA	0.09	0.08	7.43%	0.97

Table 2: Actual vs Predicted 2024 GHG/GDP

Country	Predicted 2024 GHG/GDP	Actual 2024 GHG/GDP	Error
Thailand	0.85	0.90	-0.04
Viet Nam	1.34	1.45	-0.11

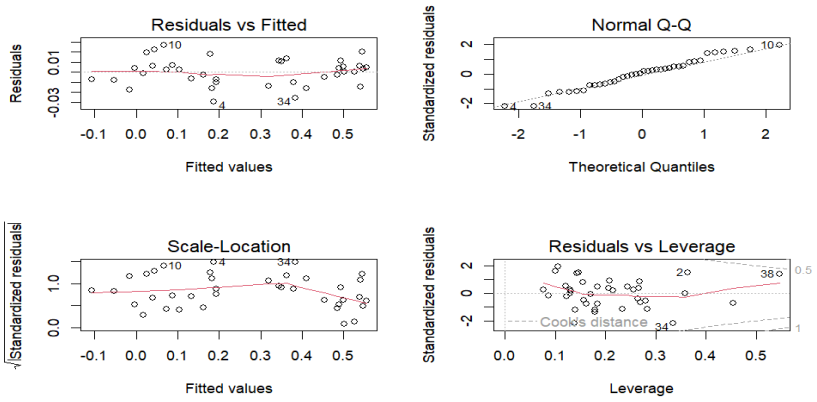


Figure 8: Diagnostic Plots

The residuals-versus-fitted plot shows residuals mostly centred around zero, with only slight curvature, suggesting no major systematic bias. The Q-Q plot indicates approximate normality in the central residuals, but some tail deviations suggest a few outlying observations. The scale-location plot shows a mild increase in residual spread at higher fitted values, indicating slight heteroscedasticity. The residuals-versus-leverage plot identifies a few higher-leverage observations, particularly observation 38, but no point appears to exceed Cook's distance substantially. Overall, the diagnostics support the use of Model 2 for forecasting.

The coefficient results provide useful implications for the 2030 stress test. The coefficient for lagged log GHG/GDP is 0.13, but it is not significant, suggesting that short-term persistence is present but not strong enough to independently explain emissions intensity after controlling for other predictors. Renewable energy has a negative and significant coefficient of -0.01, meaning that higher renewable energy consumption is associated with lower GHG/GDP intensity. Energy intensity has a positive and significant coefficient of 0.11, indicating that economies requiring more energy per unit of GDP tend to have higher emissions intensity.

Forest area has a negative and significant coefficient of -0.02, suggesting that greater forest coverage is associated with lower GHG/GDP intensity. Manufacturing has a positive coefficient of 0.01, but it is only marginally significant, so its effect should be interpreted cautiously. Year has a negative and significant coefficient of -0.01, indicating that emissions intensity has generally declined over time after controlling for energy, forest, manufacturing and country effects. Overall, the signs and significance of renewable energy and energy intensity are the most important for the later strategy proposal.

6.0 Climate Mitigation Strategy

6.1 Proposed Mitigation Strategy: Energy Transition and Efficiency Improvement

The proposed mitigation strategy is an **energy-transition and efficiency-improvement pathway**, using higher renewable energy adoption and lower energy intensity as the main levers. This aligns with the model because both variables affect projected **2030 GHG/GDP emissions intensity**. Higher renewable energy reduces fossil-fuel reliance, while lower energy intensity means less energy is needed per unit of economic output. Together, they reduce emissions pressure and support a lower climate-risk pathway.

This strategy is suitable for Thailand and Viet Nam because both countries are already moving toward cleaner energy systems. Thailand focuses on renewable power expansion and energy-efficiency improvement, while Viet Nam's PDP VIII supports a gradual shift from coal toward renewables, grid modernisation, battery storage, hydrogen, and ammonia.

From a reinsurance perspective, the reinsurer does not directly implement energy-transition policies. Instead, transition progress acts as a forward-looking climate-risk signal for later decisions on pricing, capacity, risk appetite, and risk transfer.

7.0 Stress Test and 2030 Projection Impact

7.1 Stress-Test Design and Assumptions

The stress test is conducted using a **GHG/GDP emissions intensity dashboard**, which forecasts each country's 2030 emissions intensity and tests how the forecast changes under different transition scenarios. The target variable is **GHG per billion USD GDP**, calculated as total greenhouse gas emissions excluding LULUCF divided by GDP in billions of USD. Therefore, the stress test evaluates emissions intensity rather than total emissions alone.

The dashboard first constructs a **2030 baseline forecast** using the dynamic panel fixed-effects model. Renewable energy share and energy intensity are then adjusted to test mitigation and adverse transition pathways, while other predictors remain unchanged. This allows the stress test to isolate the effect of stronger or weaker transition progress on projected 2030 emissions intensity.

The stress-test adjustment follows:

Table 3: Stress-Test Scenario Assumptions and Expected Impact

Scenario	Stress-test assumption	Expected GHG/GDP impact
Baseline 2030	Current model trend continues; no stress adjustment	Reference projection
Mitigation 2030	Renewable energy share +10% Energy intensity −10%	Lower GHG/GDP than baseline
Adverse 2030	Renewable energy share −5% Energy intensity +10%	Higher GHG/GDP than baseline

Table 4: Stress-Test Adjustment Formulae

Variable	Formula
Renewable energy share	Renewable stressed = Renewable baseline × (1 + renewable adjustment / 100)
Energy intensity	Energy intensity stressed = Energy intensity baseline × (1 + energy intensity adjustment / 100)

A negative energy-intensity adjustment represents improved energy efficiency. These shocks are used as moderate scenario adjustments, not exact forecasts. The purpose is to test whether stronger or weaker transition progress materially changes 2030 emissions intensity.

7.2 2030 Projection Impact

The stress-test results show that the mitigation scenario reduces projected 2030 GHG/GDP emissions intensity in both Thailand and Viet Nam, while the adverse scenario increases it. This indicates that higher renewable energy adoption and lower energy intensity can reduce transition-risk pressure, while weaker transition progress can worsen emissions intensity.

Table 5: 2030 GHG/GDP Projection Under Baseline, Mitigation and Adverse Scenarios

Country	Baseline 2030 GHG/GDP	Mitigation GHG/GDP	Change vs baseline	Adverse GHG/GDP	Change vs baseline
Thailand	0.7961 Mt/B$	0.7498 Mt/B$	-5.82%	0.8411 Mt/B$	+5.65%
Viet Nam	1.2479 Mt/B$	1.1799 Mt/B$	-5.45%	1.3115 Mt/B$	+5.09%

The **Transition Risk Index (TRI)** is anchored at **100** under the baseline scenario. A TRI below 100 indicates lower emissions intensity than baseline, while a TRI above 100 indicates deterioration.

Table 6: Transition Risk Index Under 2030 Stress-Test Scenarios

Country	Baseline TRI	Mitigation TRI	Adverse TRI
Thailand	100.00	94.18	105.65
Viet Nam	100.00	94.55	105.09

Under the mitigation scenario, Thailand's projected emissions intensity decreases by **5.82%**, while Viet Nam's decreases by **5.45%** relative to baseline. Under the adverse scenario, Thailand's emissions intensity increases by **5.65%**, while Viet Nam's increases by **5.09%**. These results show that transition progress has a measurable impact on projected 2030 emissions intensity. While the indicator does not directly forecast insurance claims, a worsening transition pathway may signal higher future insured-loss pressure, which can then inform reinsurance pricing, capacity and risk-transfer decisions.

8.0 Reinsurance Strategy and Financial Impact

The 2030 stress-test results are translated into a scenario-based reinsurance strategy. Since GHG/GDP emissions intensity is a forward-looking transition-risk signal, the reinsurer should not use it as a standalone pricing model. Instead, it should guide decisions on pricing margins, treaty capacity, catastrophe protection, retrocession, and portfolio monitoring. Financially, the strategy affects the reinsurer through four channels: pricing margins, capital buffer, treaty capacity, and risk-transfer cost.

The financial impact begins at the **insured-loss level**. Higher climate-risk pressure may increase natural disaster frequency and severity, raising insurance claims for catastrophe-exposed lines such as property, engineering and business interruption. However, only part of economic loss becomes insured loss, depending on insurance penetration, policy coverage, deductibles and insured asset concentration. For the reinsurer, the impact arises when these insured claims are transferred through reinsurance treaties. Therefore, GHG/GDP is not a direct predictor of

reinsured claims, but a forward-looking signal for pricing margins, treaty capacity, catastrophe protection and retrocession planning.

Table 7: Reinsurance Response and Financial Impact by Stress-Test Scenario

Scenario Result	Reinsurance Response	Financial Impact
Baseline: current pathway continues	Maintain current treaty structure, climate scenario margin, and annual monitoring of exposure, disaster losses, and accumulation risk.	Keeps existing capital buffer and reduces the risk of underpricing climate uncertainty.
Mitigation: GHG/GDP lower than baseline	Offer resilience-linked terms, stable capacity, or additional capacity for cedants with credible renewable-energy, energy-efficiency, and climate-resilience plans.	Supports more stable pricing, lower climate margins, and better capital efficiency.
Adverse: GHG/GDP higher than baseline	Increase catastrophe loading, tighten capacity, review treaty limits, and strengthen Cat XoL, aggregate stop-loss, quota-share, or retrocession protection.	Raises premium and risk-transfer cost, but protects profitability and solvency against higher climate-loss volatility.

To make the stress test financially actionable, the reinsurer can translate the Transition Risk Index into treaty-level climate margins. Under the mitigation scenario, the TRI falls to 94.18 for Thailand and 94.55 for Viet Nam, supporting stable capacity and limited additional climate loading. Under the adverse scenario, the TRI rises to 105.65 for Thailand and 105.09 for Viet Nam. As a practical scenario-based underwriting rule, a deterioration of around 5% in TRI may justify an indicative 2%–3% climate risk loading, subject to exposure quality, insured asset concentration, historical catastrophe losses and cedant retention levels. This does not imply that emissions intensity directly predicts claims, but it provides a structured and transparent way to translate climate stress results into pricing and capital decisions.

Thailand and Viet Nam require different reinsurance responses because their catastrophe risk profiles differ. **Thailand** has a low-frequency but high-severity flood-tail profile, so the reinsurer should strengthen **Catastrophe Excess-of-Loss (Cat XoL)**, review attachment points, reinstatement terms, climate-adjusted PML assumptions, and vertical retrocession. This protects the reinsurer from sudden large capital strain caused by extreme flood events.

In contrast, **Viet Nam** faces more recurring storm and flood losses, creating aggregate loss pressure across the treaty year. Therefore, the reinsurer should use **aggregate stop-loss**, quota-share support, trend loading, and annual exposure monitoring. This reduces earnings volatility from repeated medium-sized losses and helps stabilise cedant results.

This strategy also links to energy transition. Cedants with credible renewable-energy exposure, energy-efficient insured portfolios, and strong risk controls may receive better treaty terms or more capacity. Weak-transition portfolios may face higher loading, reduced capacity, or stricter treaty conditions. Hence, reinsurance becomes both a **capital-protection tool** and a **financial incentive** for cleaner and more climate-resilient portfolios.

Overall, the financial impact is reflected through pricing adequacy, capital allocation, underwriting appetite, treaty capacity, and risk-transfer cost. Under a mitigation pathway, the reinsurer can support transition through more stable pricing and capacity. Under an adverse pathway, it should protect profitability and solvency by increasing risk margins, tightening capacity, and strengthening catastrophe protection.

9.0 Conclusion

This study assessed how upstream climate-pressure indicators, particularly GHG emissions intensity, can support reinsurance risk assessment in Southeast Asia, using Thailand and Viet Nam as paired case studies. Three key insights emerge. First, the modelling results show that emissions intensity is structurally linked to energy efficiency and renewable energy adoption. Thailand follows a more stable emissions-intensity pathway, while Viet Nam shows stronger emissions growth relative to economic growth, indicating higher transition-risk pressure.

Second, Thailand and Viet Nam present different catastrophe-risk profiles. Thailand is mainly **severity-driven**, with low-frequency but high-impact flood events such as the 2011 floods dominating the loss experience. Viet Nam is more **frequency-driven**, with recurring storm and flood losses. This means Thailand requires stronger catastrophe excess-of-loss protection and flood accumulation monitoring, while Viet Nam requires aggregate loss monitoring, pricing adequacy and protection-gap management.

Third, the stress-test results show that the energy-transition pathway is financially relevant for reinsurance decisions. Under the adverse scenario, rising GHG/GDP intensity for both countries supports the need for climate loading in treaty pricing. In contrast, credible mitigation progress can support more stable capacity and resilience-linked terms.

The study has limitations. The model is based on two countries, 2030 projections rely on stable-trend assumptions, and EM-DAT captures economic losses rather than insured claims directly. Also, the stress test isolates only two transition levers: renewable energy share and energy intensity, while other factors such as GDP growth, insurance penetration, adaptation investment and catastrophe exposure are held constant. Therefore, the results should be interpreted as directional impacts on GHG/GDP emissions intensity, not precise forecasts of future insured or reinsured claims.

Overall, the findings suggest that climate science and reinsurance pricing should be integrated more closely. For Hannover Re, the immediate priority is to strengthen climate-adjusted pricing, catastrophe accumulation monitoring, reserve buffers and risk-transfer solutions that reflect both transition risk and physical catastrophe exposure.

Appendix

Interactive Stress-Test Dashboard

An interactive Shiny dashboard was developed to support the 2030 stress-test analysis. The dashboard allows users to adjust the renewable energy share and energy intensity assumptions and observe the resulting GHG/GDP emissions intensity projections for Thailand and Viet Nam under baseline, mitigation and adverse scenarios.

Dashboard link: https://loosijie.shinyapps.io/masa2/

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